Prompting is not integration.

Where an LLM goes from demo candy to a component you can actually ship.

10 slides · swipe

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A prompt is a spell. Production is plumbing.

The demo works because everything is ideal: model up, input clean, no discourse. Production is malformed JSON, retries, schema drift and three-thousand-line logs at 2 a.m.

Demos convince juniors.
Plumbing convinces users.

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The five things a prompt doesn't give you.

01 Structured output guaranteed by schema, not by hope

02 Retries with backoff — bounded, never infinite

03 Fallback when the model is down — simpler model or cached answer

04 Validation on model output before any downstream write

05 A kill switch and a way to observe cost per request

Skip any one and your LLM is a mood, not a service.

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How I wire LLM features to production.

01 Strict schema on input — reject garbage before the model

↓

02 LLM call bounded by temperature 0 + max_tokens

↓

03 Output validated against JSON Schema — fail closed

↓

04 On failure: retry once, then fallback path, then human

The model is never the last gate. Validation is.

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The metrics I wire into every LLM feature.

```
$ llm-metrics

hourly request count

schema-validation fail rate

fallback usage rate

p95 latency

cost per request
```

If the model were allowed to fail silently, none of these would exist.

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Where LLM integrations actually break.

Rate limits mid-batch — retried without idempotency, duplicated writes. 'Valid JSON, invalid sense' — passes schema, fails the human read. The provider swaps the model behind the same ID — identical prompts, different answers.

**Every 'it worked in the demo'
shows up as one of these.**

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The LLM test harness worth building.

01 Golden outputs for your 50 most common prompts

02 Regression suite against prompt changes — same inputs, same schema

03 Adversarial garbage inputs — the model must refuse, not invent

04 Cost assertion per request — budget as a test, not a hope

Without this, every model upgrade is a silent roulette spin.

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Two ways teams ship LLM features.

PROMPTING

Prompt-first

magic words
works on demos
breaks on Mondays

INTEGRATION

Systems-first

schema + retry
fallback + observe
ships without fear

One is a trick. The other is infrastructure.

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The LLM production law.

The prompt is 10% of the work. The other 90% is making the system survive without its personality.

Models have moods. Systems shouldn't.

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LLM · KEEP THIS · 10/10

Save this for your next LLM launch.

Before the first production call: schema contract, fallback path, budget alarm, regression set. Four boxes. No applause until they're checked.

Save for later review

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